

## 07 · Did the regional campaign work? — synthetic control (Anchor B)

July 11, 2026

### 1 07 · Did the regional campaign work? — synthetic control (Anchor B)

**The business decision.** We ran a TV campaign in one metro area — a **DMA** (*Designated Market Area*, the standard US media-buying region) — and left the rest of the country dark. Sales in that metro are up since launch. Before we spend to roll the campaign out nationally: **was the lift caused by the campaign, or would sales have risen there anyway?**

#### 1.0.1 Why this is hard: there was no control group

Unlike the uplift notebook (01), here there is **no randomized hold-out** — we didn’t flip a coin to decide which metro got the campaign, we just picked one. So we can’t simply compare it to “the others.” And we can’t just compare the treated metro’s sales *after* vs *before*, because sales everywhere are pushed around by a shared **trend** (secular growth), **seasonality**, and a common **macro wave** (the economy, category demand). A naive before/after would credit the campaign with an uplift the market would have delivered on its own. We need the one thing we can never observe: **what sales in that metro *would have been* if we had not run the campaign** — its *counterfactual*.

#### 1.0.2 The idea: build the counterfactual from the other markets

**Synthetic control** (Abadie & Gardeazabal, 2003) constructs that missing counterfactual as a **weighted blend of the untreated markets** — the “**donor pool**.” It searches for a set of weights so that the blend closely tracks the treated market’s sales *in the pre-launch period*. If a weighted combination of, say, 60% Denver + 25% Phoenix + 15% others matched our metro week-for-week *before* the campaign, then — assuming those markets keep moving together — that same blend is our best guess of what our metro *would* have done *after*, absent the campaign. The **gap** between the metro’s actual sales and this “synthetic” version, after launch, is the estimated lift.

The reason it can work at all: all markets share the same underlying **factors** (trend, season, macro). A blend of donors that matches the treated unit’s pre-period is, in effect, matching it on those shared factors — so the blend and the treated unit stay locked together *unless something (the campaign) breaks them apart*.

We do this the Bayesian way, so instead of one number we get a **posterior** — a full distribution of plausible lifts — which gives us honest uncertainty, a real significance test (a **placebo per-**

**mutation**, defined in Depth B), and robustness checks a VP of Growth can interrogate before committing national budget.

### 1.0.3 How this notebook is organised

The **7-step contract** (question · simulate a known truth · identify · estimate · validate · decide in € · caveats) plus the full **three depths**:

- **Depth A** — why the tempting naive estimators fail (a bake-off)
- **Depth B** — placebo inference in *space* and *time*, with the scale-free Abadie RMSE-ratio test
- **Depth C** — the euro rollout decision and its sensitivity to assumptions

FAST=False SC sampling={'draws': 1500, 'tune': 1500, 'chains': 4}

## 1.1 2 · Simulate a ground truth (why fake data first)

As in every notebook here, we **cannot grade a causal method on real data** — the counterfactual (“what our metro would have done without the campaign”) is never observed. So we build a simulator where we *plant* the true lift, confirm the method recovers it, and only then trust it.

The simulator produces a weekly **sales panel**: a table with one row per market per week, for 30 DMAs over 60 weeks. Each market’s sales are built from **shared latent factors** — a rising trend, a seasonal cycle, and a random macro wave common to all markets — combined with that market’s own baseline level, its own sensitivity to each factor (“factor loadings”), and idiosyncratic noise. The campaign launches in **week 40** in market 0 only, adding a **true +12% lift** from then on.

The shared macro wave is the villain: it moves *every* market together, so it’s exactly what a naive before/after would mistake for campaign effect. But it’s also the hero of the method — because the factors are *shared*, a weighted blend of the *other* markets can reconstruct market 0’s untreated path. That is precisely what makes the effect *identifiable* here (recoverable in principle), and the plot below shows the setup: one dark line (treated) inside a bundle of grey donors, all riding the same wave.

**The data-generating model** — exactly what `dgp.geo_panel` implements (defaults & seed in `src/cmp/dgp.py`). Markets  $i = 0, \dots, 29$ , weeks  $t = 0, \dots, 59$ . Three **shared latent factors**:

$$g_t = 0.4 t \quad (\text{trend}), \quad s_t = 8 \sin\left(\frac{2\pi t}{26}\right) + 4 \sin\left(\frac{2\pi t}{13}\right) \quad (\text{season}), \quad m_t = \sum_{u \leq t} \eta_u, \quad \eta_u \sim \mathcal{N}(0, 1.2^2) \quad (\text{macro random})$$

Each market has its own baseline  $c_i \sim U(80, 140)$  and factor loadings  $\lambda_i = (\lambda_{i1}, \lambda_{i2}, \lambda_{i3})$ , each  $\sim U(0.6, 1.4)$ :

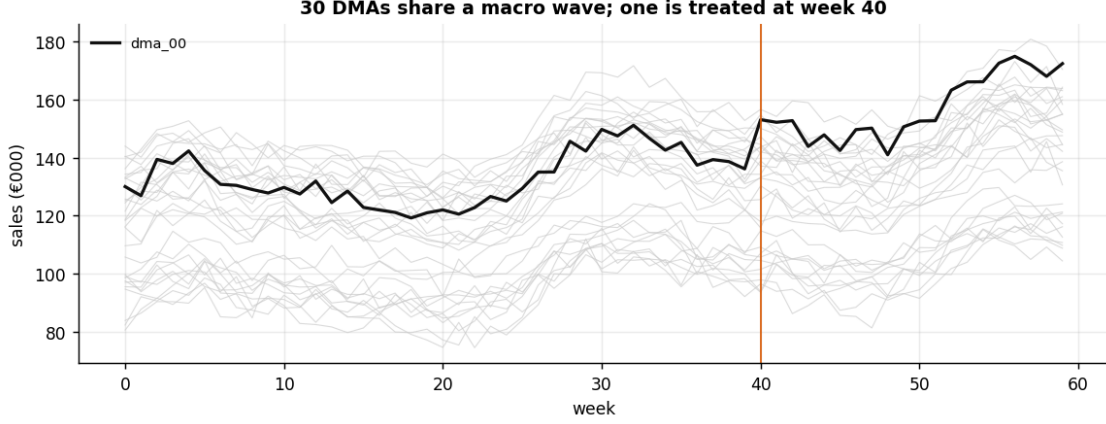
$$Y_{it} = c_i + \lambda_{i1} g_t + \lambda_{i2} s_t + \lambda_{i3} m_t + \varepsilon_{it}, \quad \varepsilon_{it} \sim \mathcal{N}(0, 3^2),$$

and the campaign adds a **proportional lift to market 0 only**, from week 40:

$$Y_{0t} = c_0 + \lambda_{01} g_t + \lambda_{02} s_t + \lambda_{03} m_t + \varepsilon_{0t} + \Delta_t, \quad \Delta_t = 0.12 (c_0 + \lambda_{01} g_t + \lambda_{02} s_t + \lambda_{03} m_t) \mathbf{1}[t \geq 40].$$

This factor structure is precisely why synthetic control works here: every market rides the *same*  $(g_t, s_t, m_t)$ , only with different loadings, so a convex combination of donors whose blended loadings match  $(\lambda_{01}, \lambda_{02}, \lambda_{03})$  reproduces market 0's untreated path. It's also why the naive estimators fail — the random-walk  $m_t$  moves all markets together, and a before/after comparison books that shared drift as “lift.”

Treated: dma\_00, launch week 40. TRUE lift €17.0k/week, total €339k.



## 1.2 3 · Identify — the estimand and the assumptions it rests on

**The estimand (the precise thing we *want*).** For each post-launch week  $t$ , the causal effect on the treated market is the potential-outcome contrast

$$\tau_t = Y_{1,t}(1) - Y_{1,t}(0), \quad t > T_0,$$

where  $Y_{1,t}(1)$  is **observed** (the campaign ran) but  $Y_{1,t}(0)$  — what week- $t$  sales *would have been* with no campaign — is the **missing counterfactual**.

**The estimator.** Synthetic control fills in that counterfactual with a weighted average of donor markets, so the effect we actually compute is

$$\hat{\tau}_t = Y_{1,t} - \underbrace{\sum_j w_j Y_{j,t}}_{\widehat{Y_{1,t}(0)} = \text{synthetic control}}, \quad t > T_0,$$

where  $Y_{1,t}$  is the **treated** market's sales in week  $t$ ,  $Y_{j,t}$  is **donor** market  $j$ 's sales,  $T_0$  is the launch week, and  $w_j$  are the **donor weights**. The weights are chosen to match the treated market's *pre-launch* path and are constrained to the **simplex**: each  $w_j \geq 0$  and they sum to 1 ( $\sum_j w_j = 1$ ). Think of the simplex constraint as “the synthetic market must be a genuine *weighted average* of real markets” — no negative weights, no scaling beyond the observed range.

**Why the simplex matters (the core of Abadie's method).** If instead we let weights be any numbers (ordinary regression), the “synthetic” market could sit *outside* the range of any real

market — **extrapolation** dressed up as a match, which fits the pre-period beautifully and then diverges wildly after. Forcing weights onto the simplex keeps the synthetic inside the **convex hull** of the donors (loosely, “inside the cloud of real markets”), which is what makes the post-launch projection credible.

**The assumptions — each gets an explicit check later:**

| assumption                            | plain meaning                                                                                      | checked in                                                                                                                                                                                 |
|---------------------------------------|----------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| <b>convex hull / no extrapolation</b> | the treated market lies inside the donor cloud; simplex weights enforce it                         | weights (Step 4)                                                                                                                                                                           |
| <b>good pre-fit</b>                   | the synthetic actually tracks the treated market <i>before</i> launch                              | pre-RMSE gate (Step 5)                                                                                                                                                                     |
| <b>no anticipation</b>                | nothing shifts sales <i>before</i> launch (no leaked campaign, no pre-buying)                      | placebo-in-time (Depth B)                                                                                                                                                                  |
| <b>stable factor structure</b>        | donors and treated keep co-moving <i>after</i> launch (the loadings matched pre-launch still hold) | long pre-period + placebo-in-time                                                                                                                                                          |
| <b>no spillover / interference</b>    | the campaign in market 0 doesn’t change donor markets (SUTVA for geos)                             | assumed by design (simulation); on real data, exclude adjacent / media-overlapping donors and re-run — placebo-in-space <i>cannot</i> detect spillover (it contaminates every placebo too) |
| <b>no concurrent treated shock</b>    | nothing else hit market 0 exactly at launch                                                        | untestable — institutional knowledge only                                                                                                                                                  |

### 1.3 4 · Estimate — a Bayesian synthetic control

We put a **Dirichlet prior** on the donor weights and fit them to the pre-period. (The *Dirichlet* is the natural probability distribution over the simplex — draws from it are automatically non-negative and sum to 1 — so it’s the Bayesian way to say “the weights live on the simplex.”) Fitting returns a **posterior** over the weights, and therefore a posterior over the whole counterfactual path and the lift — Abadie’s constrained least squares, but with uncertainty for free.

**The fitted model, in symbols.** With  $Y_{0t}$  the treated market and  $Y_{jt}$  the  $J = 29$  donors, fit **on the pre-period only** ( $t < 40$ ):

$$Y_{0t} \sim \mathcal{N}\left(\sum_{j=1}^J w_j Y_{jt}, \sigma^2\right), \quad w \sim \text{Dirichlet}(\mathbf{1}_J), \quad \sigma \sim \text{HalfNormal}(5).$$

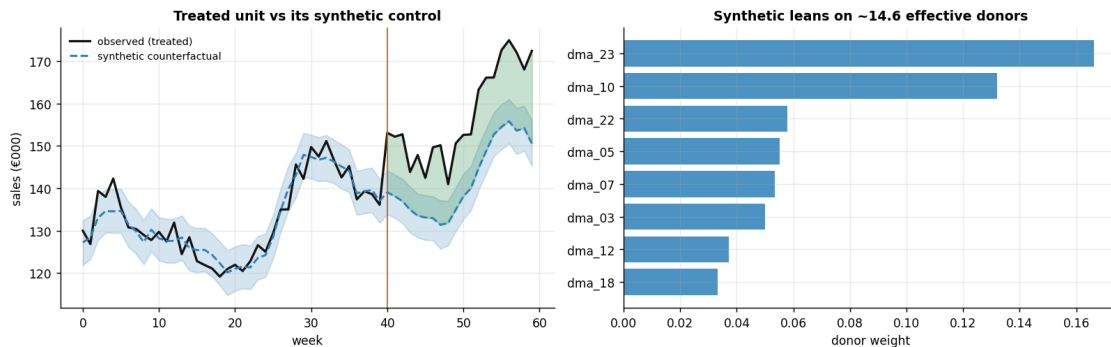
$\text{Dirichlet}(\mathbf{1}_J)$  is flat over the simplex — no donor favoured a priori, weights non-negative and summing to 1 by construction. The counterfactual for *every* week is then the posterior-predictive

blend  $\hat{Y}_{0t}^{(0)} = \sum_j w_j Y_{jt} + \varepsilon_t$  with  $\varepsilon_t \sim \mathcal{N}(0, \sigma^2)$  — the  $\varepsilon_t$  matters: dropping it (using the mean blend alone) understates the uncertainty band, which is exactly the coverage lesson step 5 quantifies — and the weekly lift is  $Y_{0t} - \hat{Y}_{0t}^{(0)}$ .

Two things to read off the fit: - **Which donors** the synthetic leans on (the weight bars). One caveat to state plainly: the Dirichlet prior produces a *regularized blend* spread over many donors, rather than classical synthetic control’s sparse handful. That’s **more stable** out-of-sample but **less individually interpretable**. We summarise the spread as an “**effective number of donors**”  $= 1/\sum_j w_j^2$  (the inverse Herfindahl index — 1 if all weight is on one donor,  $J$  if spread evenly over  $J$  donors). - **Pre-fit RMSE** — the root-mean-square gap between the treated market and its synthetic *before* launch, in €000. Small means the synthetic is a good stand-in; this is the precondition for believing the post-launch gap — Step 5 turns this into a concrete PASS/FAIL gate (pre-RMSE below ~a third of the weekly lift it must detect).

**On real data.** You don’t need our simulator — this method runs on *your own* weekly (or daily) sales-by-market panel, which every company already has: one treated market, the rest as donors. The canonical *public* example is Abadie’s **California Proposition 99** tobacco study (California vs 38 other US states); it has the identical shape (one treated unit, many donors, a known intervention date) and is the standard dataset to reproduce this on.

SC sampling convergence: max r-hat 1.000 - min ESS 3994 - divergences 0  
pre-fit RMSE €2.86k · estimated total lift €305k (true €339k) · 90% CI [€270k, €339k] - at full precision the truth sits just OUTSIDE this band (\$5's multi-seed check explains why the total-lift band runs optimistic)



## 1.4 5 · Validate — recovery, calibration, the pre-fit gate, and robustness

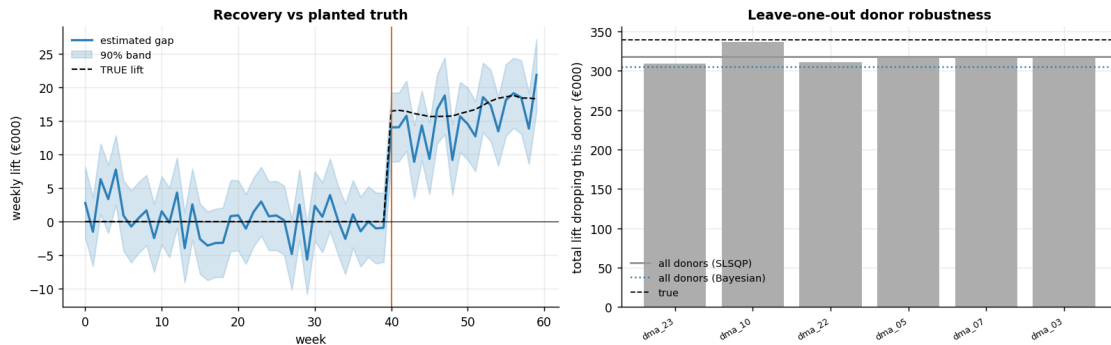
Four checks — and one of them delivers a genuinely humbling result we keep rather than hide:

1. **Recovery & calibration.** The true per-week path should sit inside the posterior band. That band is a **posterior predictive**: it carries not only *which-donors* (weight) uncertainty but the treated market’s own idiosyncratic noise. That noise term is what makes the per-week band near-nominal; a weights-only band is too narrow and can leave the truth *outside* a “90%” interval.
2. **Multi-seed calibration.** One panel is a single draw of the world, so we refit on fresh panels

and report the recovery bias and coverage. The honest result: the **per-week** band is near-nominal, but the **total-lift** interval covers well *below* nominal — synthetic control’s genuine sample-to-sample variance is larger than any single panel’s posterior width. So the posterior interval is an *optimistic* guide to the total, and the trustworthy inference is the **placebo permutation test** (design-based, calibrated by construction), not the band. This is *why* the euro call below is “confirm with a second test,” not “the interval says go.” (*The per-seed refits in this check are deliberately coarse 300-draw probes, read only in aggregate; their sampler chatter is silenced.*)

3. **Pre-fit gate** — the pre-period gap must hover around **zero**: concretely, pre-RMSE below about a third of the weekly lift the design must detect (printed as an explicit PASS/FAIL below). A bad synthetic match makes the post gap meaningless — this is the precondition, not a formality.
4. **Leave-one-out donor robustness** — drop each of the top-weighted donors and refit; if the estimate lurches when one donor leaves, the result hangs on a single market and is fragile.

pre-fit gate: pre-RMSE €2.86k vs  $|weekly\ lift|/3 = €5.1k \rightarrow$  PASS  
pre-launch gap mean €0.30k ( $\sim 0$  = good fit). L00 total range €309-337k vs the same-fitter SLSQP baseline €318k (Bayesian €305k) – no single donor drives it.



Across 24 fresh panels – recovery bias €+23k (+7%), with a LARGE seed-to-seed sd of €59k.

PER-WEEK band covers the true path 86% of the time – ~ nominal; the posterior-predictive obs-noise term earns this.

But the TOTAL-lift 90% interval covers only ~50% across panels: SC's sample-to-sample variance dwarfs any single panel's posterior width, so the total interval is optimistic.

=> the CALIBRATED inference is the placebo permutation test (below), not the posterior band alone.

## 1.5 6 · Decide, in euros

The campaign cost **€300k**. Everything the notebook has produced so far now collapses into a single object: the **posterior over total incremental sales** across the 20 post-launch weeks (**total\_post**, in €000). The rollout question is *not* “is the lift positive?” — Depth B’s placebo permutation answers that — but “**what is the probability the lift beats what we paid?**”, because a real-but-too-small effect still loses money at national scale.

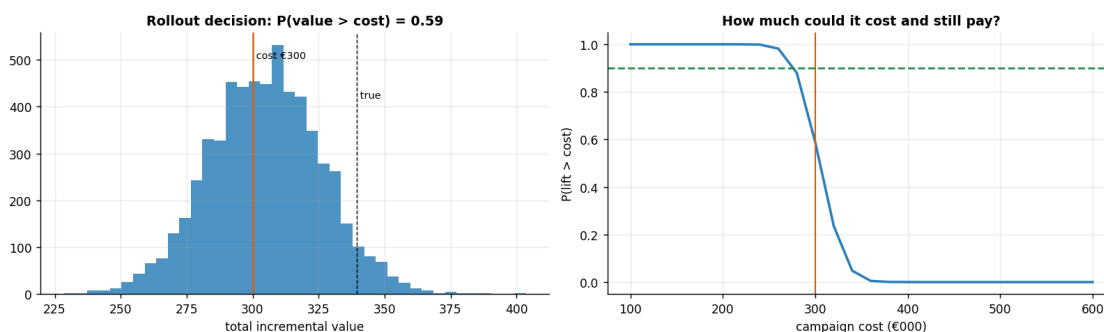
**The decision rule** (`policy.go_no_go`, the cookbook’s standard): read  $P(\text{value} > \text{cost})$  off the posterior and act on a stated convention —

- $P > 0.9 \rightarrow$  **GO** — sure enough to act without buying more evidence;
- $P < 0.5 \rightarrow$  **NO-GO** — more likely than not the campaign doesn’t pay at this price;
- in between  $\rightarrow$  **TEST FURTHER** — the honest middle where the effect may well be real but the *economics* are not yet settled.

Alongside the probability it reports the expected **net value** ( $\mathbb{E}[\text{value}] - \text{cost}$ , in €000) and the expected **ROI** as a ratio ( $\text{net} \div \text{cost}$ , so break-even is 0) — three views of the same posterior, so nobody has to reverse-engineer the call from a chart.

**How to read the two panels.** *Left:* the posterior of total lift with the €300k cost line — the mass to the right of the line is  $P(\text{lift} > \text{cost})$  (the dashed marker is the true total, which we only know because this is a simulation). *Right:* the same posterior read as a **headroom curve** — sweep a hypothetical cost  $c$  and plot  $P(\text{lift} > c)$ . Where the curve crosses the green 0.9 line is the *highest price at which this evidence would already justify a confident GO* — the number to bring to the media-buying negotiation, not just a yes/no.

```
{
  "P_value_gt_cost": 0.5878333333333333,
  "expected_value": 304.7713414404065,
  "expected_net_value": 4.771341440406502,
  "expected_roi": 0.015904471468021673,
  "value_lo": 270.2408765084909,
  "value_hi": 339.44759544458793,
  "cost": 300.0,
  "decision": "TEST FURTHER"
}
```



**Read-out.** The verdict in this run is **TEST FURTHER**, and the JSON says exactly why: the expected lift ( $\sim$  €305k) sits almost on top of the €300k cost — expected net value  $\sim$  **€5k**, ROI  $\sim$  **+1.6%** — and  $P(\text{lift} > \text{cost}) \approx 0.59$ , barely better than a coin flip. Hold this against Depth B without flinching at the apparent contradiction: the placebo permutation says the effect is **real** ( $p \sim 0.03\text{--}0.05$ ), while the economics say it is **marginal at this price**. Both are true at once — “statistically significant” and “worth the money” are different questions, and conflating them is how marginal campaigns get scaled. The headroom curve prices the gap: the 0.9 crossing sits

at ~ €275k in this run, so had the campaign cost ~8% less, this same evidence would already be a confident GO. At €300k, the evidence buys conviction that *something* happened, not conviction that it *paid*.

**A borderline call has a price tag: the value of information.** “TEST FURTHER” must not end as a shrug — the deliverable is an answer to “*what evidence would settle this, and is it worth buying?*” Two tools, both computed below:

- **VOI (value of information).**  $\text{VOI} = \mathbb{E}[\max(V - c, 0)] - \max(\mathbb{E}[V] - c, 0)$  — the expected gain from deciding *after* the uncertainty resolves versus committing now on the mean. Intuition: if the truth lands below the cost line, perfect foresight would have let us walk away and keep the difference; VOI is that averted loss, averaged over the posterior. It is the most a *perfectly informative* test could add in expected decision value — i.e. an **upper bound on any confirmatory-test budget**.
- **Test sizing.** A second geo test treating  $k$  independent markets tightens the posterior roughly as  $1/\sqrt{k}$  — it **sharpens** the estimate around the same mean; it doesn’t move it. So “how big must the test be?” becomes: *the smallest  $k$  whose tightened posterior reaches  $P(\text{lift} > \text{cost}) \geq 0.9$* .

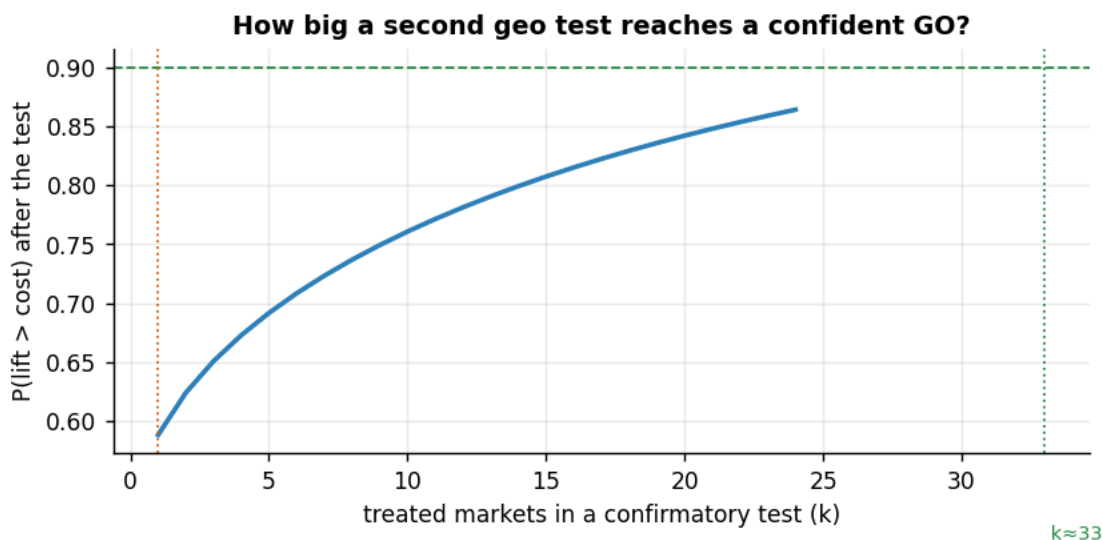
Value of information: resolving the cost-line uncertainty is worth ~€6k of expected decision value.

Reaching  $P(\text{lift} > \text{cost}) \geq 0.90$  would take ~33 independent treated markets - more than this

world contains (30 DMAs, one already treated). The honest read: NO feasible single confirmatory

test gets to 0.90 here; extend the post-period, pool repeated tests, or accept a lower confidence bar -

and the multi-seed check says even this arithmetic is optimistic. Still an experiment, never a blind launch.



**Read-out — and an honest dead end.** VOI ~ €6k in this run: with the posterior centred



almost exactly on the cost line, even perfect information barely changes the expected outcome — whichever way the truth falls, we're near break-even, so the *stakes of deciding wrong are small*. That immediately disciplines the next step: a confirmatory test is only rational if it costs less than ~€6k of all-in decision value — no €50k research program is justified by these economics. The sizing arithmetic is blunter still: reaching the 0.9 bar would take ~ **33 independent treated markets, more than this world contains** (30 DMAs, one already treated). So no feasible single geo test settles the question at this cost. The honest options, in order: **renegotiate the price** (at ~ €275k the existing evidence already clears 0.9), **extend the post-period or pool repeated tests** (more weeks per market buys precision too), or **accept a lower confidence bar** knowing the multi-seed check in step 5 says these intervals are already optimistic. What is *not* on the list: a blind national rollout. Depth C pulls this thread together with the launch-date and donor-pool sensitivities into the one-paragraph verdict a VP signs.

## 1.6 7 · Caveats

- **Here synthetic control is high-variance, not conservative.** In this panel the point estimate lands ~10% low, but across seeds (Step 5) it shows only a small mean bias (+7% in our run) but a *large* sample-to-sample spread — and its posterior interval is *optimistic* about the total lift. So trust the **placebo permutation test**, not the point estimate or the band.
- **Pre-fit is a gate, not a formality.** High pre-RMSE => the post gap is uninterpretable; no Bayesian machinery fixes a bad donor pool.
- **No-anticipation & no-spillover are assumptions**, stress-tested by the placebos in Depth B.
- **One treated unit = limited power.** Significance comes from the permutation test, coarse with few donors.

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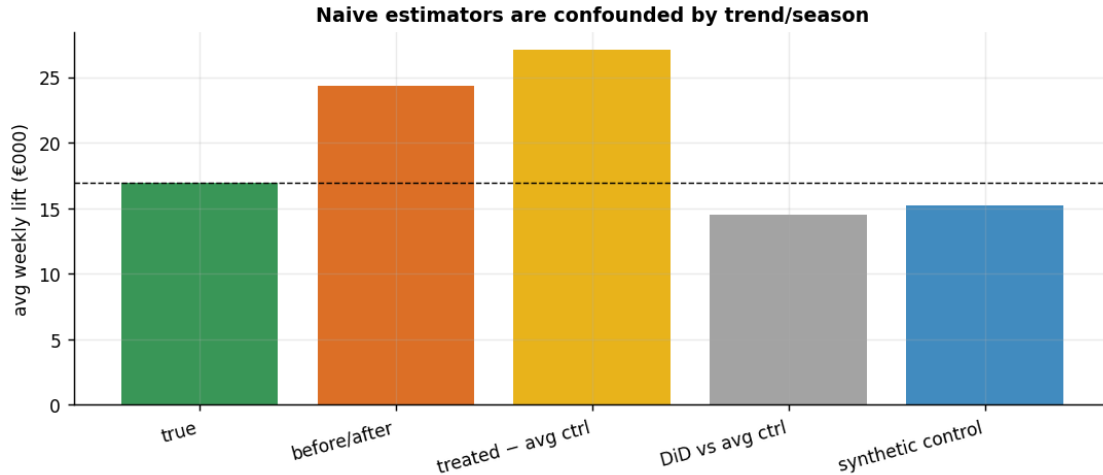
## 2 Depth A · Why the naive estimators fail

Two tempting shortcuts, both confounded by the shared trend/season/macro wave:

- **Before/after:** post mean — pre mean of the treated DMA — books the entire trend as effect.
- **Treated — average control:** the average control has the wrong factor loadings, so it doesn't cancel the common wave cleanly.

Synthetic control is the fix: a *weighted* control chosen to match pre-launch.

```
{
  "true": 17.0,
  "before/after": 24.3,
  "treated \u2212 avg ctrl": 27.1,
  "DiD vs avg ctrl": 14.5,
  "synthetic control": 15.2
}
```



**Read-out.** Before/after badly overstates the lift (it books the trend); treated-minus-average is **worse** — it books the raw between-market level gap, differencing nothing out; differencing out the shared trend (DiD vs the average control) removes the trend bias yet a *level* gap from mismatched factor loadings remains; **the reweighted synthetic control comes closest** (€15.2k vs the true €17.0k — an 11% undershoot; DiD-vs-average lands at €14.5k, and neither naive shortcut is even close). Same data, four estimates — the estimator *is* the identification strategy.

### 3 Depth B · Placebo inference — is the gap statistically real?

We have a lift estimate, but with **one treated market** there's no classical standard error to test it (standard errors assume many independent units). Instead we ask a **permutation** question, the inferential heart of synthetic control:

*If the campaign had no effect, how unusual would a post-launch gap this big be, just by chance?*

**Placebo-in-space.** We answer it by pretending, in turn, that *each donor market* was the treated one: fit that donor its own synthetic control from the remaining donors, and record its post-launch “gap.” Since none of the donors actually got a campaign, these placebo gaps map out the distribution of gaps we’d see from noise alone. If the *real* treated market’s gap sits far out in the tail of that placebo distribution, a fluke is unlikely. Two refinements from Abadie:

- **Discard placebos with a poor pre-fit** — a donor the synthetic can’t match before launch produces a meaningless gap that would pollute the comparison.
- **Use the RMSE-ratio statistic**, not the raw gap: the *post*-period RMSE divided by the *pre*-period RMSE. This is **scale-free** (fair across big and small markets) and, crucially, doesn’t reward a placebo that simply fits badly *everywhere* (high post *and* high pre  $\rightarrow$  ratio  $\sim 1$ ).

The **p-value** is then the treated market’s rank in the ratio distribution: the fraction of markets (placebos + treated) whose ratio is at least as extreme as the treated market’s. A small p — say

$\leq 0.05$  — means “only this market, out of all of them, shows a jump this sharp relative to its own pre-fit.”

placebo-in-space kept 18/29 donors (dropped 11: pre-RMSE > 5x the treated fit)

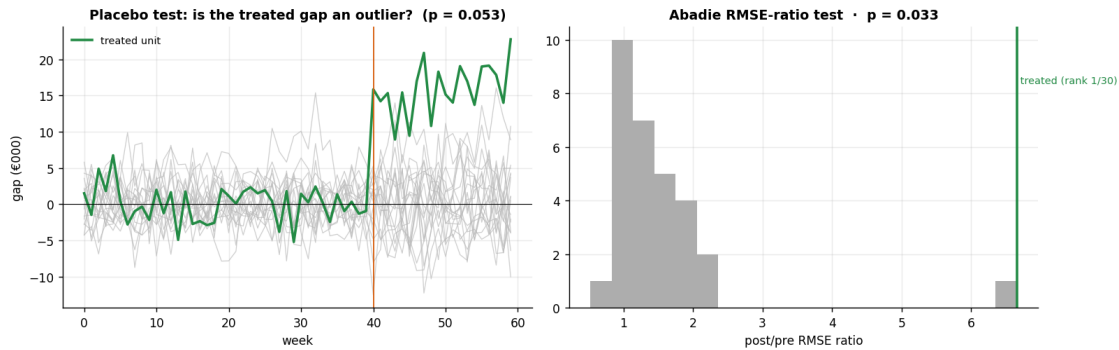
placebo p vs filter cutoff: 2x  $\rightarrow$   $p=0.056$  · 5x  $\rightarrow$   $p=0.053$  · 20x  $\rightarrow$   $p=0.333$

(tighter filters compare only well-matched placebos and sharpen the test; a loose 20x filter admits poor synthetics and dilutes it)

placebo p (gap) = 0.053 [add-one permutation, treated excluded from placebos] ·

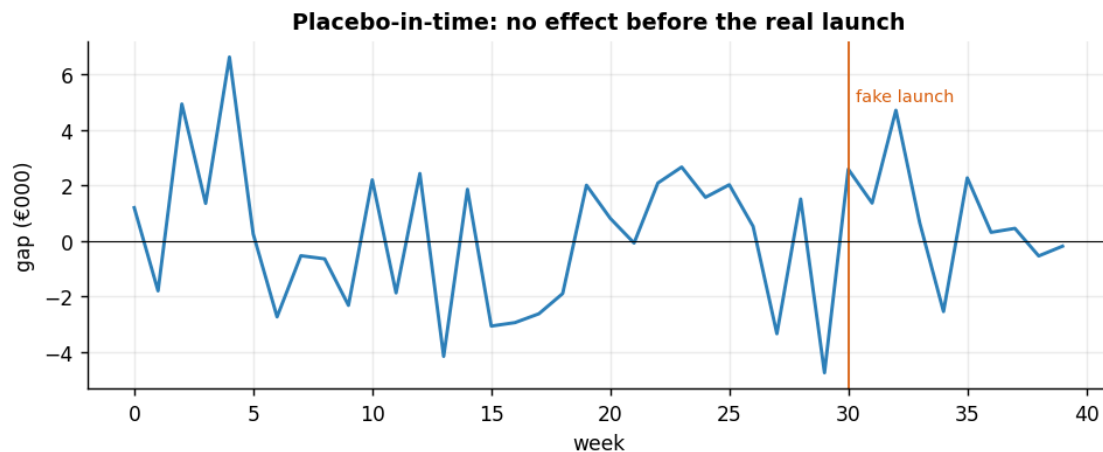
RMSE-ratio p = 0.033 [Abadie rank/N, treated included] (treated ratio 6.7 vs

placebo median 1.3; both rank the treated 1st)



**Placebo-in-time.** A second falsification: pretend the campaign launched *before* it did (a fake week inside the pre-period). The estimated “effect” should be  $\sim 0$  — a non-zero fake effect would mean the method invents lift (anticipation or overfitting).

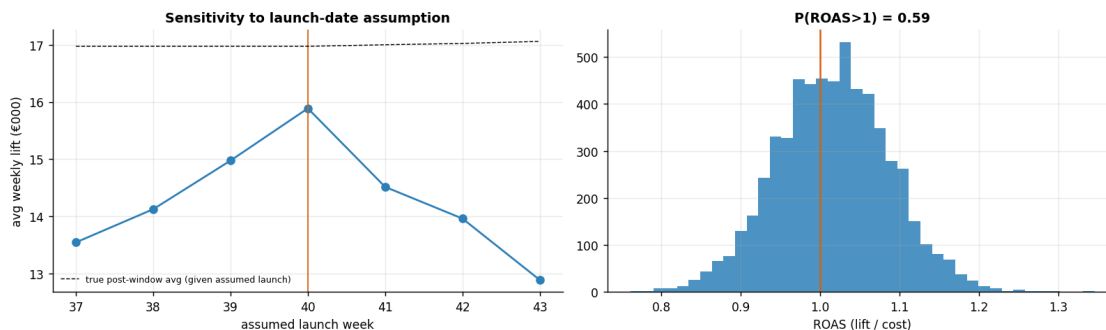
placebo-in-time avg gap €0.9k vs pre-fit RMSE €2.86k (should be  $\sim 0$ , and it is well within the pre-fit noise) - no spurious pre-effect.



## 4 Depth C · The euro rollout decision & sensitivity

Pull it together into the number a VP signs off on, and pressure-test how the call moves with the **launch-timing** assumption (did we mis-date the campaign?) and the **donor pool**.

```
{
  "true_total": 339.4806333100736,
  "estimated_total": 304.7713414404065,
  "ci90": [
    270.2408765084909,
    339.44759544458793
  ],
  "pre_rmse": 2.855007246428569,
  "p_space": 0.05263157894736842,
  "p_rmse_ratio": 0.03333333333333333,
  "placebo_in_time": 0.9046168542429143,
  "P_lift_gt_cost": 0.5878333333333333,
  "decision": "TEST FURTHER",
  "eff_donors": 14.579714199645526
}
```



### 4.0.1 The one-paragraph decision

The campaign drove a **real** incremental lift: the synthetic tracks the treated market closely before launch (low pre-RMSE, pre-gap  $\sim 0$ ), spreads its weight across  $\sim 15$  effective donors (no single market dominates), and it is robust to dropping any single donor (leave-one-out stays close to the same-fitter baseline). Nudging the *assumed* launch date does shift the estimate — but that is mechanical, not fragility: mis-dating folds no-effect weeks into the post window and dilutes the average lift, so the peak at the true launch week is itself a diagnostic. The post gap is an outlier on both the raw-gap and the scale-free RMSE-ratio **placebo permutation** tests, and the time placebo shows no spurious pre-effect — that permutation test is the calibrated evidence the effect is real. The euro call is deliberately cautious: the point estimate carries meaningful seed-to-seed variance (this panel lands  $\sim 10\%$  below truth; across panels the mean bias is small ( $+7\%$ ) but the variance is *large*), and the multi-seed check shows the **total-lift interval is optimistic** — so we do **not** let a single posterior interval authorise a na-

tional rollout. Against the €300k spend the effect sits near break-even: “**promising, confirm with a second geo test,**” not “**scale nationally today.**” The effect is clearly real; whether it clears *this* cost is the close call — and one an optimistic interval should make us *more*, not less, cautious about.